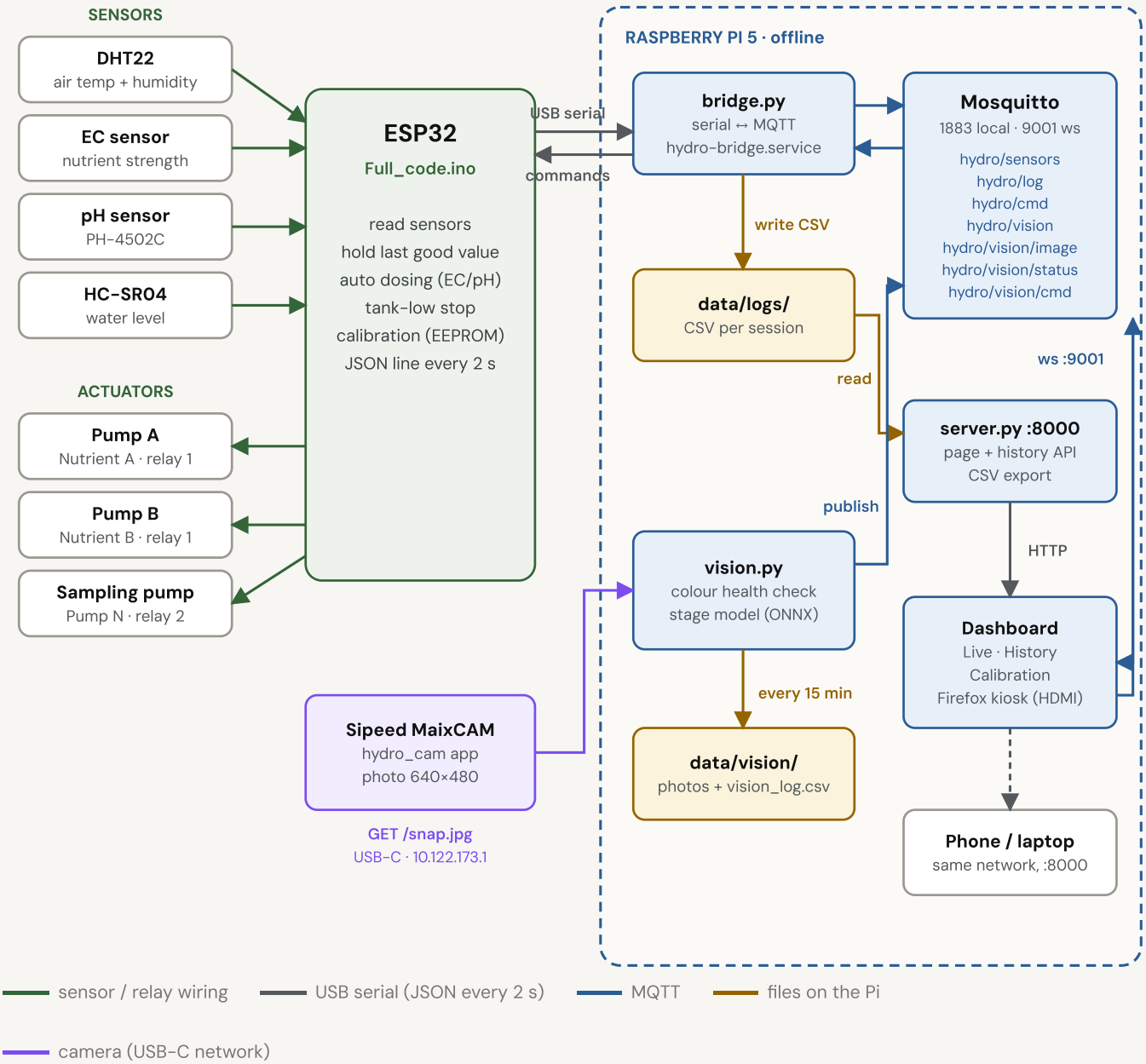
04

System Diagram & Flowchart

How data moves from the sensors through the ESP32, the USB cable and the MQTT broker to the logs, the camera AI and the dashboard, and how the firmware decides when to run the pumps.

1 · ARCHITECTURE

Block diagram



In short: the ESP32 is the only part that touches the hardware and it keeps dosing and protecting the pumps even if the Pi is off. The Pi adds memory (CSV logs), eyes (camera + AI) and a screen (dashboard). All links are local: a USB cable to the ESP32, USB-C to the camera, and MQTT on the Pi itself.

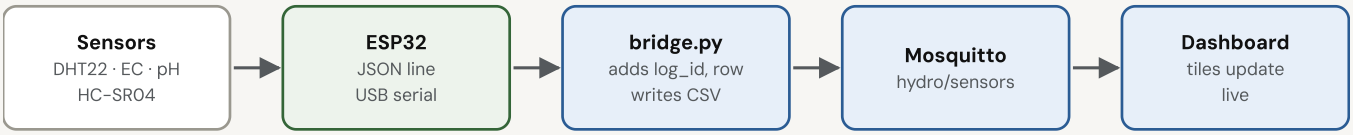
2 · MESSAGING

MQTT topics

Mosquitto runs on the Pi. Programs on the Pi use port 1883 (localhost only); the dashboard in the browser uses the WebSocket listener on port 9001.

Topic	Published by	Read by	Content
`hydro/sensors`	bridge.py	Dashboard	One JSON reading every 2 s (+ `log_id`, `row`)
`hydro/log`	bridge.py	Dashboard	ESP32 text messages (calibration replies, warnings)
`hydro/cmd`	Dashboard, `mosquitto_pub`	bridge.py → ESP32	Commands: `status`, `enterec`, `ph7`, `pumpauto`, …
`hydro/vision`	vision.py	Dashboard	Health, canopy, yellowing, stage, confidence, size (JSON)
`hydro/vision/image`	vision.py	Dashboard	Latest photo as JPEG bytes (retained)
`hydro/vision/status`	vision.py	Dashboard	`online` or `offline: <reason>`
`hydro/vision/cmd`	Dashboard	vision.py	`snap` = save a photo now

Path of one sensor reading



Path of a command (e.g. calibration)



Services on the Pi (start-up order)

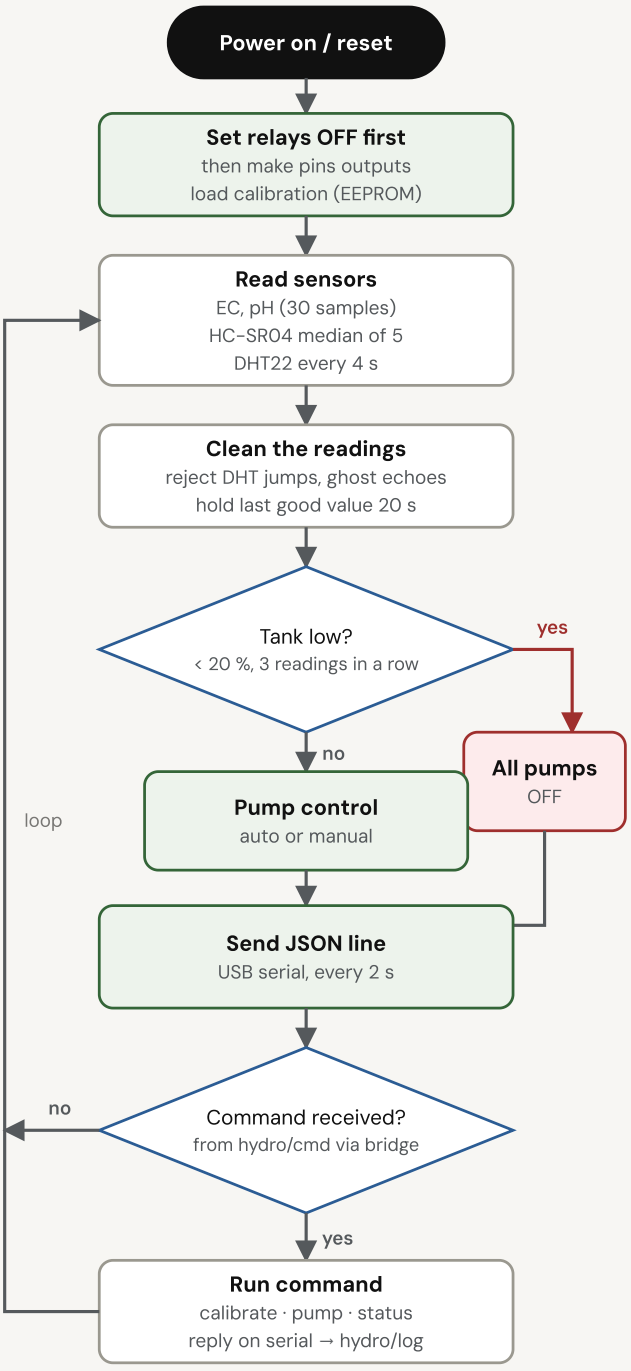
Service	Starts when	Runs
`mosquitto`	Boot	MQTT broker (1883 local, 9001 websocket)
`hydro-bridge`	ESP32 plugged in (udev → `/dev/esp32`); stops when unplugged	`backend/bridge.py`
`hydro-vision`	Boot	`backend/vision.py`, photo every 10 s
`hydro-dashboard`	Boot	`backend/server.py` on port 8000
`hydro-kiosk` (user)	Desktop log-in	Firefox full screen on the HDMI monitor, screen never blanks

> Everything runs offline: the dashboard's libraries (`mqtt.js`, uPlot) and fonts are stored in `frontend/vendor/`. If the browser cannot reach port 9001, the camera photo still updates over HTTP from `frontend/live/`.

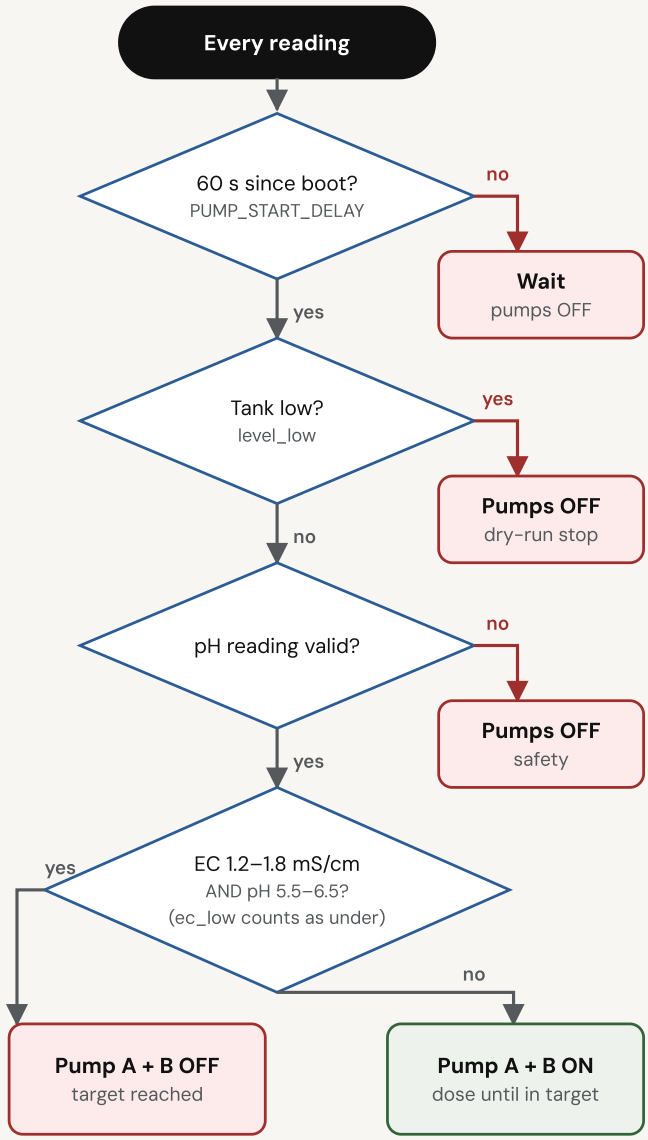
3 · FLOWCHARTS

ESP32 firmware logic

Main loop



Pump A + B decision (auto mode)



Sampling pump

starts 30 s after boot, then always on
stops only when the tank is low
or on the sampleoff command

Camera check (vision.py, every 10 s)

